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September 30, 2019
NRC-19-0071

10 CFR 50.73

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2019-004

Pursuant to 10 CFR 50.73(a)(2)(v)(D), DTE Electric Company (DTE) is submitting LER No. 2019-004, "Division 2 Mechanical Draft Cooling Tower Fan "D" Tripped when Placed in High Speed Due to a Blown Fuse."

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Jason R. Haas, Manager – Nuclear Licensing, at (734) 586-1769.

Sincerely,

A handwritten signature in blue ink, appearing to read "P. Fessler".

Paul Fessler
Senior Vice President and CNO

Enclosure: Licensee Event Report No. 2019-004, "Division 2 Mechanical Draft Cooling Tower Fan "D" Tripped when Placed in High Speed Due to a Blown Fuse."

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov)

**Enclosure to
NRC-19-0071**

**Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2019-004
Division 2 Mechanical Draft Cooling Tower Fan "D" Tripped when
Placed in High Speed Due to a Blown Fuse**



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Fermi 2					2. Docket Number 05000 341			3. Page 1 OF 3		
4. Title Division 2 Mechanical Draft Cooling Tower Fan "D" Tripped when Placed in High Speed Due to a Blown Fuse										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
07	30	2019	2019	004	00	09	30	2019	N/A	05000
9. Operating Mode			11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
			☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
			☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
			☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
10. Power Level			☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
100			☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
			☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
			☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
			☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
					☐ 50.73(a)(2)(i)(C)		☐ Other (Specify in Abstract below or in NRC Form 366A)			
12. Licensee Contact for this LER										
Licensee Contact Fermi 2 / Jason R. Haas - Manager, Nuclear Licensing								Telephone Number (Include Area Code) (734) 586-1769		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES	
X	BS	FU	N/A	Y	N/A	N/A	N/A	N/A	N/A	
14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No										
Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)										
On July 30, 2019, at 1014 EDT, the Division 2 Mechanical Draft Cooling Tower (MDCT) Fan "D" was declared inoperable due to a trip of the fan while placing it in high speed. The MDCT Fans are required to support operability of the Ultimate Heat Sink (UHS). The UHS is required to support operability of the Division 2 Emergency Equipment Cooling Water (EECW) system. The EECW system cools various safety related components, including the High Pressure Coolant Injection (HPCI) system room cooler. An unplanned HPCI inoperability occurred based on a loss of the HPCI Room Cooler. The cause of the event was a blown high speed breaker control power fuse. The control power fuse was replaced, the MDCT Fan "D" was tested satisfactorily, and HPCI was declared operable on July 30 at 1431 EDT. The blown fuse was quarantined and will be sent for failure analysis. The remaining fuses in the system are being replaced and sent to the vendor for comparison with the blown fuse.										



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Fermi 2	05000- 341	YEAR 2019	SEQUENTIAL NUMBER 04	REV NO. 00

NARRATIVE

INITIAL PLANT CONDITIONS

Mode 1

Reactor Power 100%

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On July 30, 2019, at 1014 EDT, the Division 2 Mechanical Draft Cooling Tower (MDCT) [CTW] Fan [FAN] "D" was declared inoperable due to a trip of the fan when placing it in high speed during a Technical Specification Surveillance test. The MDCT Fans are required to support operability of the Ultimate Heat Sink (UHS) [BS]. The UHS is required to support operability of the Division 2 Emergency Equipment Cooling Water (EECW) system. The EECW system cools various safety related components, including the High Pressure Coolant Injection (HPCI) [BJ] system room cooler [CLR]. An unplanned HPCI inoperability occurred based on a loss of the HPCI Room Cooler. Investigation revealed that a high speed breaker control power fuse [FU] had blown. The control power fuse was replaced, the MDCT Fan "D" was tested satisfactorily, and the UHS and HPCI were declared operable on July 30, 2019 at 1431 EDT.

The HPCI system is a single train safety system.

The 10CFR50.72(b)(3)(v)(D) notification was made at 1542 on July 30, 2019 to the NRC Operations Center by Event Notification 54188.

This 10CFR50.73(a)(2)(v)(D) notification is fulfilling the 10CFR50.73 report requirement. Based on the equipment history of the MDCT fans and the cause of the failure being a blown fuse, there is no firm evidence to identify that the failure would have existed previously. Therefore, the failure is assumed to have occurred at the time of its discovery based on the guidance in NUREG-1022 Section 3.2.2.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences associated with this event. At no time during this event was there a potential for endangering the public health and safety. There were no radiological releases associated with this event.

Technical Specification 3.7.2 requires the UHS reservoir to be operable in Modes 1, 2, and 3. The UHS operability requirements in Modes 4 and 5 are determined by the systems the UHS supports. Each reservoir is the cooling source for that division's service water subsystems. A two-cell MDCT is located above each of the one-half capacity reservoirs. Each cell is equipped with a MDCT fan. Two MDCT fans above each one-half capacity reservoir are required for that division of UHS to be considered operable. The "A" and "C" MDCT fans are in Division 1 and the "B" and "D" fans are in Division 2.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Fermi 2	05000- 341	YEAR 2019	SEQUENTIAL NUMBER 004	REV NO. 00

NARRATIVE

Since the Division 1 MDCT fans were unaffected, Division 1 of UHS and all of its supported equipment, including RCIC, were operable throughout this event. As a result, there was no loss of safety function associated with the UHS itself. The mission time associated with HPCI is less than 24 hours. An engineering analysis has shown that use of a single MDCT fan in one division can achieve acceptable results for a 24 hour period for that division. The failure of MDCT Fan "D" only occurred at high speed and did not impact its ability to operate at low speed. The conditions present during this event, where Division 2 UHS had one operable MDCT fan (Fan "B") and a second MDCT fan operating at low speed (Fan "D") were less challenging than the conditions utilized in the engineering analysis. Therefore, the short period of time where MDCT Fan "D" was only capable of operating at low speed would have had little or no impact on the functionality of HPCI.

CAUSE OF THE EVENT

The fuse in location EF between points 1 and 2 for the MDCT Fan "D" high speed breaker 72 ED position 2B blew when the fan was shifted to high speed during the TS surveillance.

CORRECTIVE ACTIONS

The fuse was replaced. The fan was then satisfactorily tested and the 52XX relay was confirmed to be energized following the test. The 52XX relay performed as expected for MDCT Fan "D" high speed breaker.

The blown fuse was quarantined and will be sent for failure analysis. The remaining fuses in the system are being replaced and will be sent to the vendor for comparison with the blown fuse.

PREVIOUS OCCURRENCES

There have been no previous similar occurrences.